ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 0.68 m and 3 m	lambda = v/f.
3	**(c)** 24 m/s	0.6 x 40.
4	**(d)** 150 m	1500 x 0.2/2.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 2 s	680/340.
7	**(a)** 86 m	vt.
8	**(a)** 340 m/s	34/0.1 = 340 - note: for illustration.
9	**(a)** Both A and R are true and R is the correct explanation of A.	With no particles to vibrate, no compressions and rarefactions can form.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance = 170 m (1); v = 170/0.5 = 340 m/s (1).	1+1
12	Different harmonic content (1); Waveform shapes differ (1).	1+1
13	Sound travels faster in steel (1); It reaches the ear earlier (1).	1+1
14	t steel = 0.2 s (1); t air = 2.94 s; gap 2.74 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wavelengths 4 : 1 (1); Speeds equal (1); Loudness equal; pitch higher for 400 Hz (1).	1+1+1
16	T = 0.00143 s (1); lambda = 0.486 m (1); Waves in 5 s = 3500 (1).	1+1+1
17	Reflections from the walls prolong it (1); Reverberation (1); No reflections in the open (1).	1+1+1
	OR: d1 = 340 m (1); d2 = 510 m (1); Distance between cliffs = 850 m (1).	1+1+1
18	Initial distance 510 m (1); New distance 425 m (1); New echo time 2.5 s (1).	1+1+1
	OR: Bell rings but no sound in vacuum (1); Microphone converts sound to electrical signals (1); Speaker converts back (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sound Navigation And Ranging (1). (ii) $1500 \times 0.8 = 1200$ m (1). (iii) Depth $= 1200/2 = 600$ m (1). (iv) Depth $= 1500 \times 1.2/2 = 900$ m (1); ultrasound travels as a narrow beam over long distances and is not audible to humans or disturbing to sea life (any one reason) (1).	1+1+1+2
20	Observer sees flash/clap and starts stopwatch (1); Stops when the sound is heard (1); v = 1020/3 = 340 m/s (1); Repeat and take the mean (1); Precautions: large distance, wind, reaction time (1).	1+1+1+1+1
	OR: SONAR: depth and fish (1); Sonography: foetus, liver (1); Testing: cracks in metals (1); Limitations: absorption and resolution (1); Safer than X-rays (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.

- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.
- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 20 m	t1 = sqrt(2h/g), t2 = h/v: h = 20 gives 2 + 0.06.
2	**(c)** the speed of sound in steel is greater	Faster in solids.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(c)** 3.13 s	3 s + 0.13 s.
5	**(d)** 2 : 1	f = v/lambda.
6	**(c)** 1478 m/s	340/0.23.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(d)** 3 m	lambda = v/f.
9	**(c)** A is true but R is false.	A is true; loudness depends on amplitude, pitch on frequency, so R is false.
10	**(d)** A is false but R is true.	Sound is faster in steel than in air, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	d = vt/2 (1); = 300 m (1).	1+1
12	d = 340 x 8 = 2720 m (1); Light takes about 10 microseconds (1).	1+1
13	Speed is higher in helium (1); Same cavity size, so higher frequency (1).	1+1
14	Distance is small (1); Echo returns within 0.1 s and merges with the original (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Depth = 2772 m (1); Shoal depth = 693 m (1); Height above the bed = 2079 m (1).	1+1+1
16	20 Hz: 1200 per minute; 20 kHz: 1200000 (1); lambda 17 m and 1.7 cm (1); Range of hearing (1).	1+1+1
17	d1 = 1540 x 10 x 10^-6/2 = 7.7 mm (1); d2 = 9.24 mm (1); Thickness = 1.54 mm (1).	1+1+1
	OR: lambda1 = 1.344 m (1); lambda2 = 1.367 m (1); Frequency unchanged (1).	1+1+1
18	Cooler air near the ground (1); Sound refracts down (1); Less noise in the background (1).	1+1+1
	OR: In speaking, vibrations reach the ear through the bones of the skull as well (1); Recording carries sound only through air (1); Different transmission changes the quality (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $T = \frac{1}{f} = \frac{1}{500} = 0.002$ s (1). (ii) $v = f\lambda = 500 \times 0.68 = 340$ m/s (1). (iii) Yes, since 500 Hz lies within the audible range 20 Hz to 20 000 Hz (1). (iv) $\lambda = v/f = 340/1000 = 0.34$ m (1); the pitch becomes higher because the frequency is higher (1).	1+1+1+2
20	Flute: high pitch, steady (1); Drum: low pitch, decays (1); Harder strike increases amplitude (1); Loudness increases, pitch unchanged (1); Quality differs due to overtones (1).	1+1+1+1+1
	OR: d1 = 340 x 0.6/2 = 102 m (1); d2 = 170 m (1); Distance between the walls = 272 m (1); At the middle each wall is 136 m away (1); Echo time = 272/340 = 0.8 s (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.

- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.
- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** 345.6 m/s	f lambda.
3	**(d)** 1 m	v/f.
4	**(d)** the medium is removed	Sound needs a medium.
5	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes sound but does not explain the light's nature.
6	**(d)** 2 : 1	lambda proportional to 1/f.
7	**(d)** at the upper limit of human hearing	Limit.
8	**(a)** 1000 Hz	v/lambda.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Distance $= \frac{1}{2} \times 340 \times 0.1 = 17$ m.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; the frequency range defines ultrasound but is not the reason the echo method gives depth.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	T = 0.00227 s (1); lambda = 0.77 m; 880 waves (1).	1+1
12	In water 1.875 m (1); In air 0.425 m; ratio 4.4 (1).	1+1
13	lambda = 5.7 mm (1); Short wavelength detects small insects and gives directional waves (1).	1+1
14	Initial distance 34 m (1); New 17 m; time 0.1 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No medium (1); Light and radio waves carry information (1); Sound needs particles (1).	1+1+1
16	Pulse reflects from the back (1); t = 2d/v (1); d = 5000 x 4 x 10^-6/2 = 0.01 m (1).	1+1+1
17	Frequency ranges (1); Sources: human voice, earthquakes, bats (1); Uses: communication, prediction, imaging (1).	1+1+1
	OR: Table vibrates with the fork (1); A larger surface sets more air in vibration (1); Louder sound for a shorter time (1).	1+1+1
18	Air: 3 s (1); Steel: 0.204 s (1); Gap 2.8 s (1).	1+1+1
	OR: Intensity falls as 1/d squared (1); Doubling distance gives one-fourth intensity (1); Amplitude decreases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Ultrasonic (1). (ii) Its frequency is above the upper limit of human hearing, 20 000 Hz (1). (iii) $\lambda = \frac{v}{f} = \frac{340}{25000} = 0.0136$ m (1.36 cm) (1). (iv) Any two: SONAR, detecting flaws in metal castings, ultrasound scanning in medicine, cleaning delicate parts (1+1).	1+1+1+2
20	Time = 34/340 = 0.1 s (1); lambda = 5.7 mm (1); Distance from time delay (1); Size from the strength of echo (1); Direction from the two ears (1).	1+1+1+1+1
	OR: Values 331, 337, 343, 349, 355 (1); Straight line graph (1); Slope 0.6 m/s per degree Celsius (1); Intercept 331 m/s (1); Used in echo calculations (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.

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